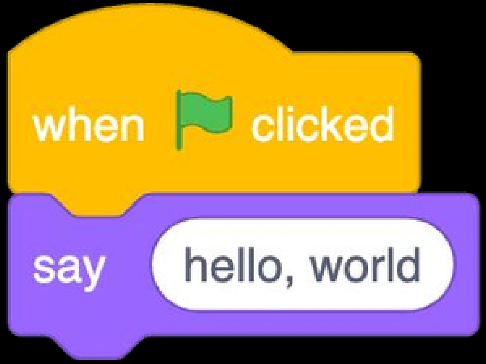
C


when clicked
say
hello, world

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

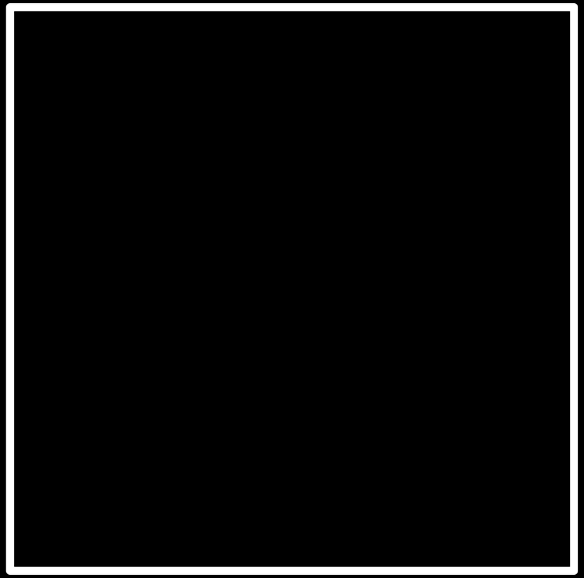
```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

código fonte

01111111 01000101 01001100 01000110 00000010 00000001 00000001 0000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 0000000
00000010 00000000 00111110 00000000 00000001 00000000 00000000 0000000
10110000 00000101 01000000 00000000 00000000 00000000 00000000 0000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 0000000
11010000 00010011 00000000 00000000 00000000 00000000 00000000 0000000
00000000 00000000 00000000 00000000 01000000 00000000 00111000 0000000
00001001 00000000 01000000 00000000 00100100 00000000 00100001 0000000
00000110 00000000 00000000 00000000 00000101 00000000 00000000 0000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 0000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 0000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 0000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 0000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 0000000
00001000 00000000 00000000 00000000 00000000 00000000 00000000 0000000
00000011 00000000 00000000 00000000 00000100 00000000 00000000 0000000
00111000 00000010 00000000 00000000 00000000 00000000 00000000 0000000

...

código de máquina

input →  → output

código fonte →

código fonte → → código de máquina

código fonte → compilador → código de máquina

Visual Studio Code

VSCode

interface gráfica do usuário

graphical user interface

GUI

interface de linha de comando

command-line interface

CLI

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```
code hello.c

make hello

./hello
```

Linux

```
cd

cp

ls

mkdir

mv

rm

rmdir

...
```


when clicked
say
hello, world

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```


say
hello, world


```
print (                    )
```


```
printf(                    )
```


```
printf( hello, world   )
```


```
printf("hello, world  ")
```


```
printf("hello, world\n")
```


```
printf("hello, world\n");
```

escape sequences

```
\n

\r

\"

\'

\\

...
```

tipos

```
bool

char

double

float

int

long

string

...
```

```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

operadores

=

<

<=

>

>=

==

!=

...

+

-

*

/

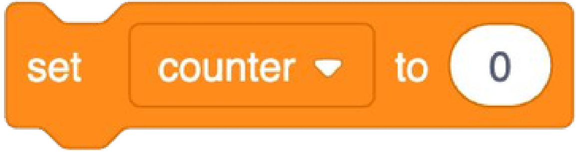
%

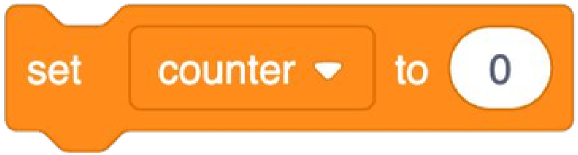
…

códigos de formatação

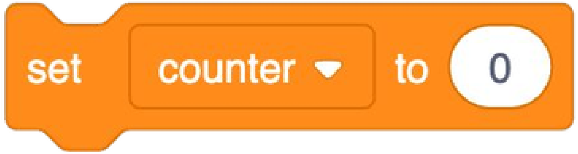
```
%c

%f

%i

%li

%s
```

variáveis

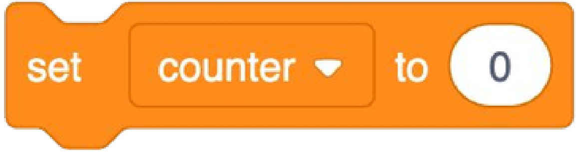

set
counter
to
0



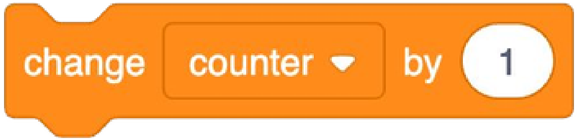

```
counter = 0
```

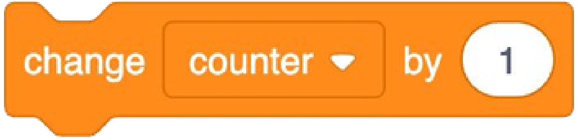



```
int counter = 0
```

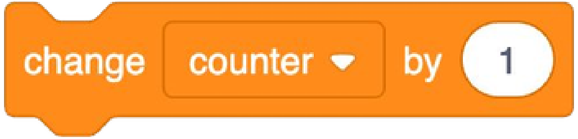



```
int counter = 0;
```

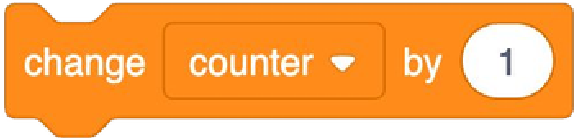

change counter by 1



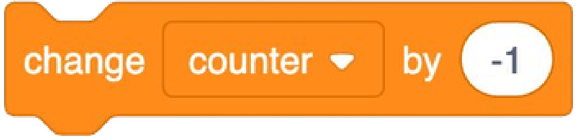

```
counter = counter + 1;
```




```
counter += 1;
```




```
counter++;
```




```
counter--;
```

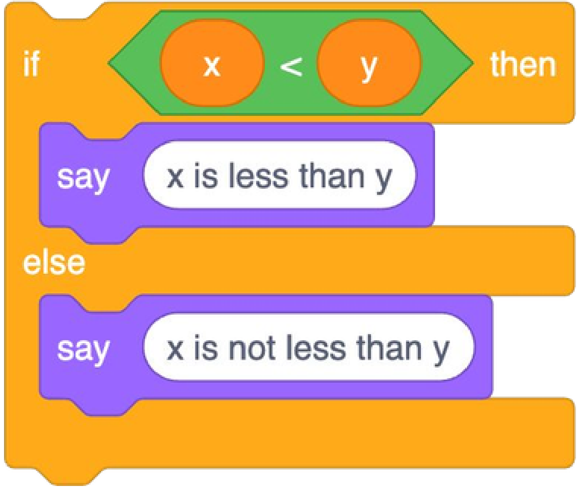
constantes

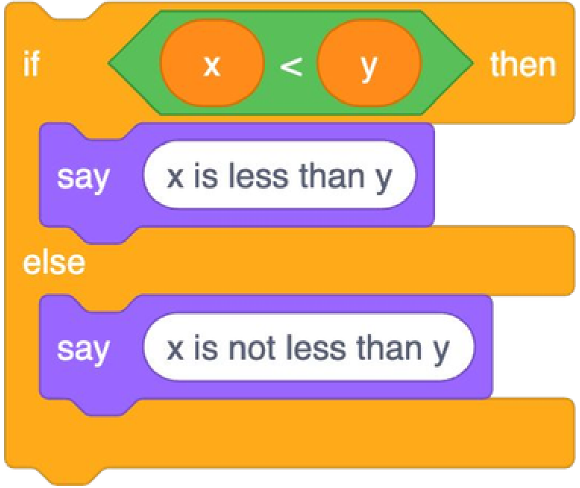
condicionais


if
x
<
y
then
say
x is less than y

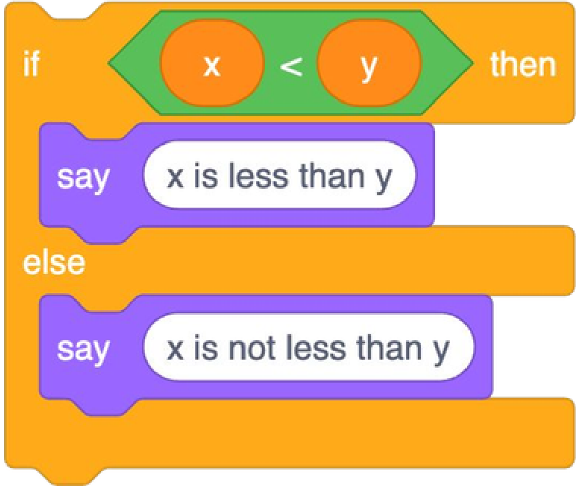

```
if (x < y)
{

}
```


```
if (x < y)
{
    printf("x is less than y\n");
}
```

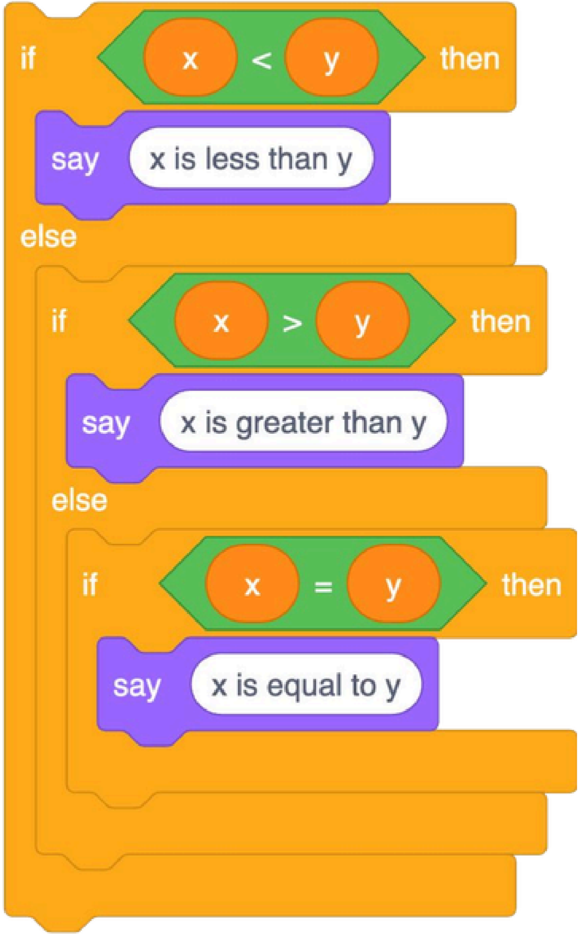

if
x
<
y
then
say
x is less than y
else
say
x is not less than y

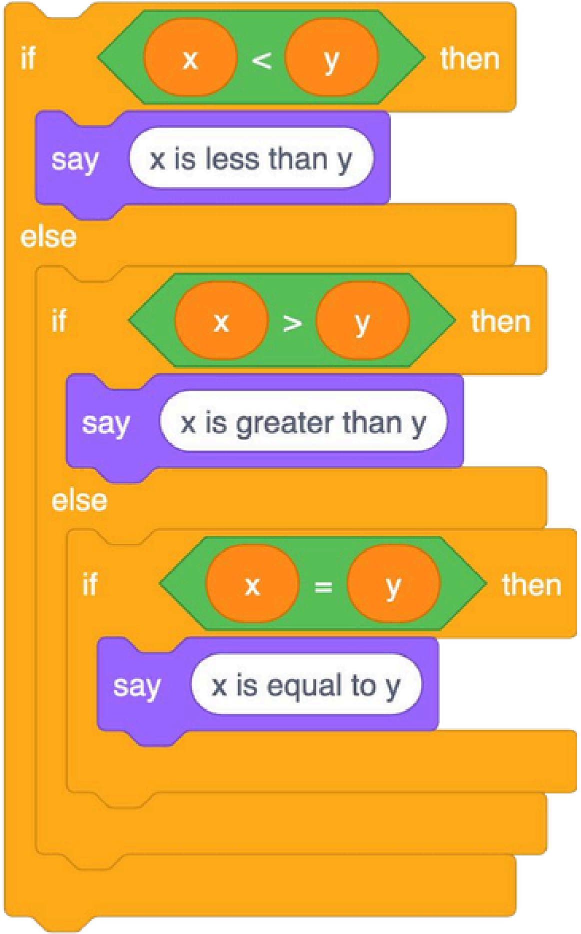



```
if (x < y)
{

}
else
{

}
```

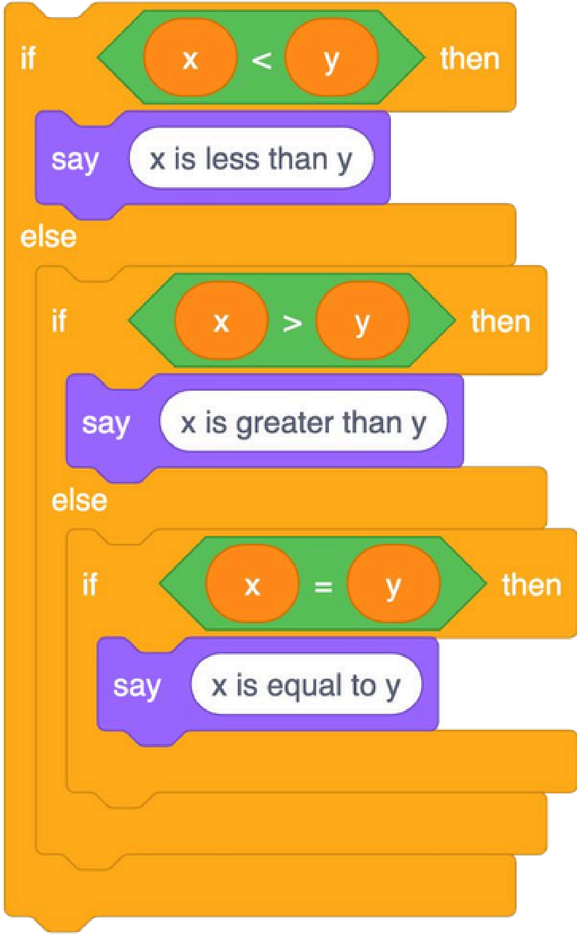



```
if (x < y)
{
    printf("x is less than y\n");
}
else
{
    printf("x is not less than y\n");
}
```

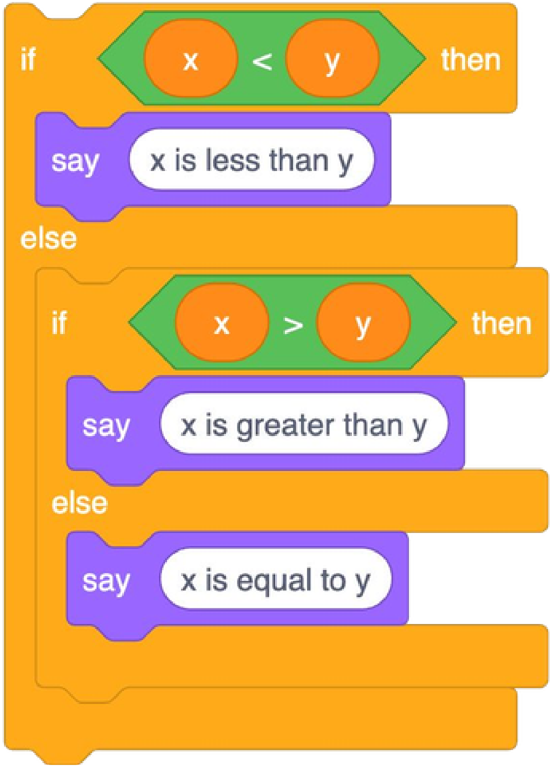

if x < y then
say x is less than y
else
if x > y then
say x is greater than y
else
if x = y then
say x is equal to y




```
if (x < y)
{

}
else if (x > y)
{

}
else if (x == y)
{

}
```




```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else if (x == y)
{
    printf("x is equal to y\n");
}
```




```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else
{
    printf("x is equal to y\n");
}
```

grafo de fluxo de controle

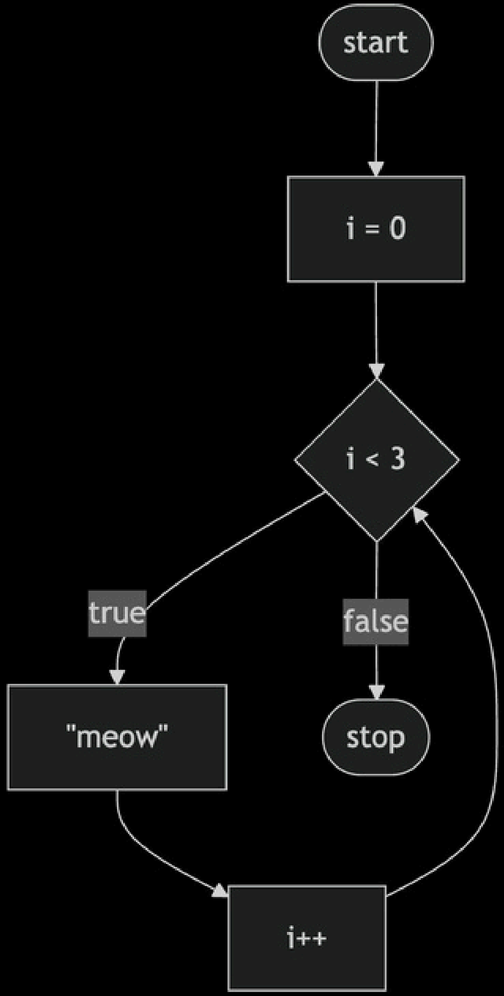
loops


repeat 3
say meow

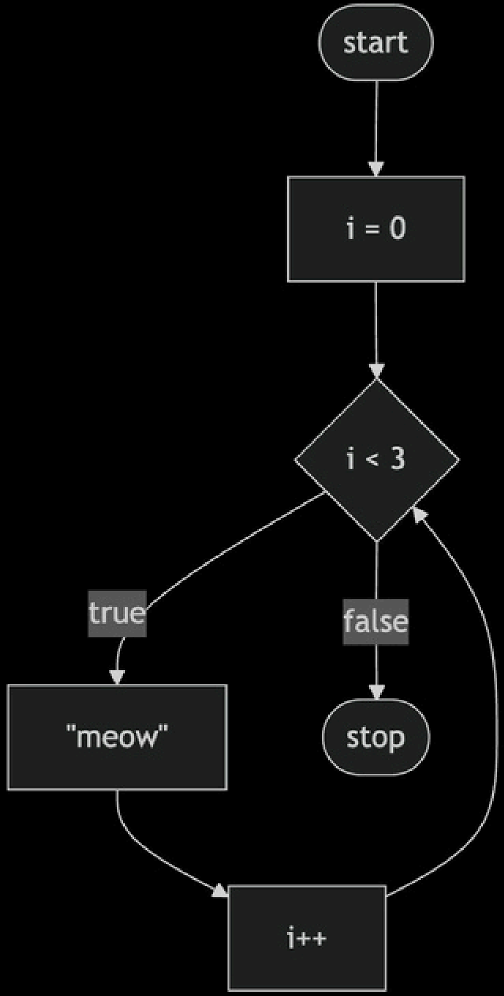

```
int i = 3;
while (i > 0)
{

    i--;
}
```


```
int i = 3;
while (i > 0)
{
    printf("meow\n");
    i--;
}
```

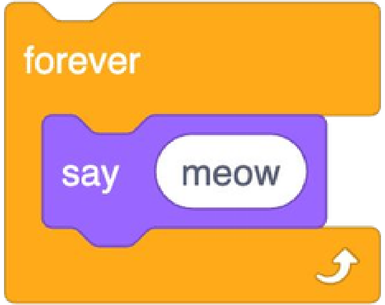

```
int i = 0;
while (i < 3)
{
    printf("meow\n");
    i++;
}
```


start
i = 0
i < 3
true
false
"meow"
stop
i++

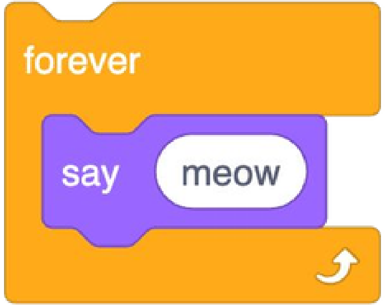

```
for (int i = 0; i < 3; i++)
{
    printf("meow\n");
}
```


start
i = 0
i < 3
true
false
"meow"
stop
i++

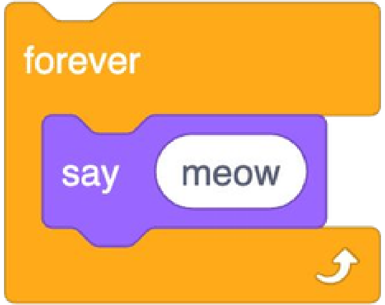

forever
say
meow



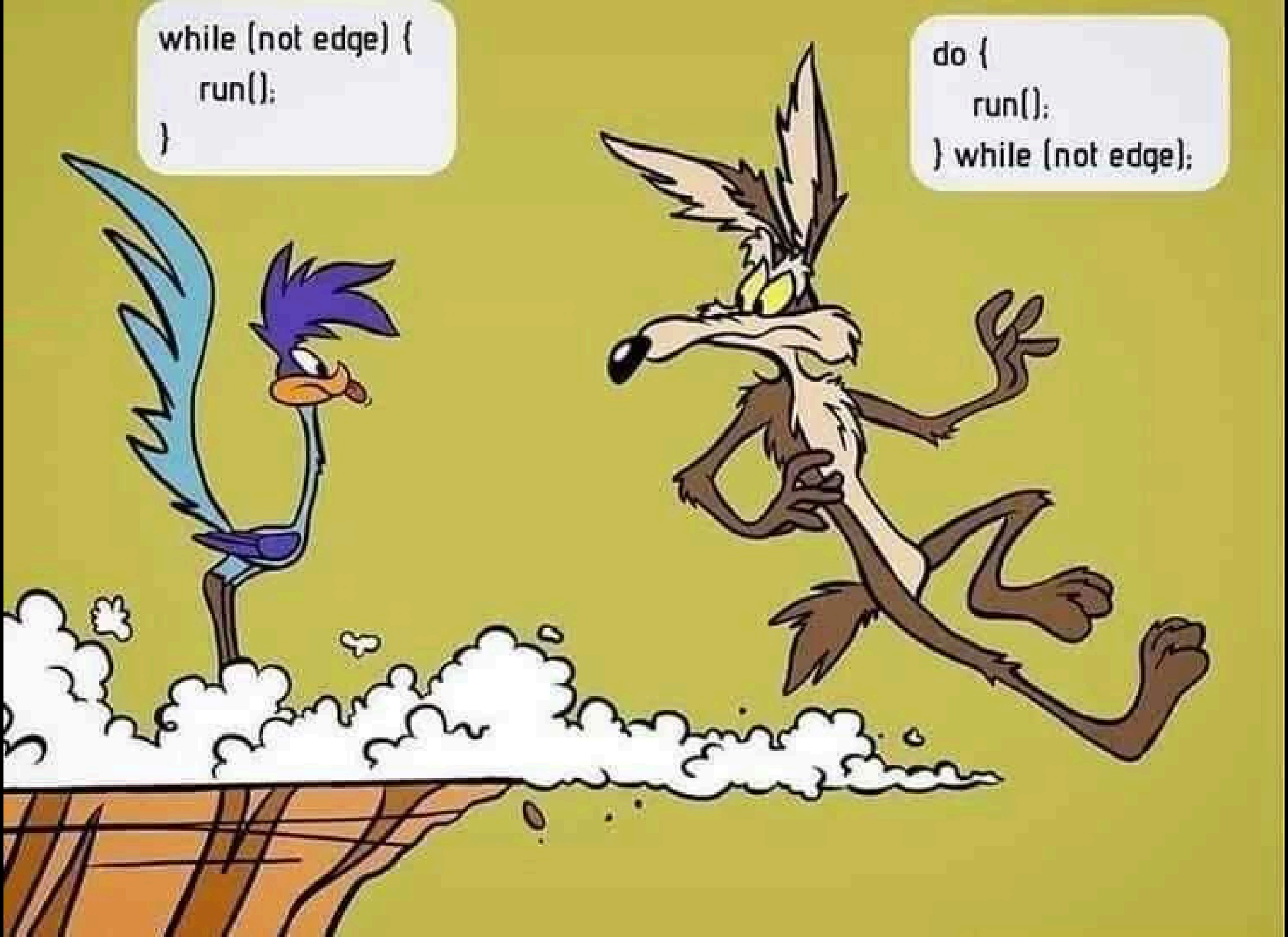

```
while(      )
{

}
```

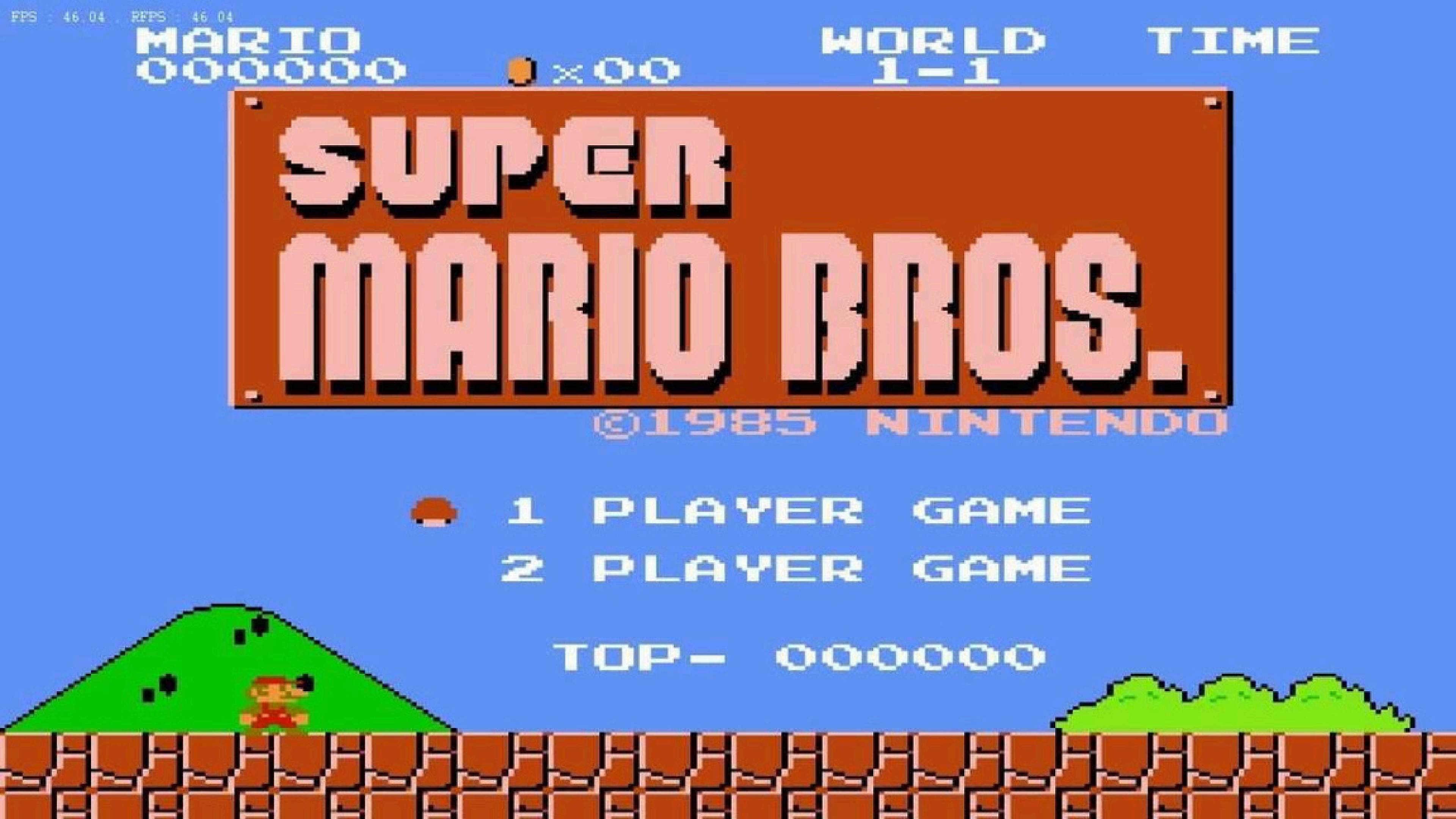


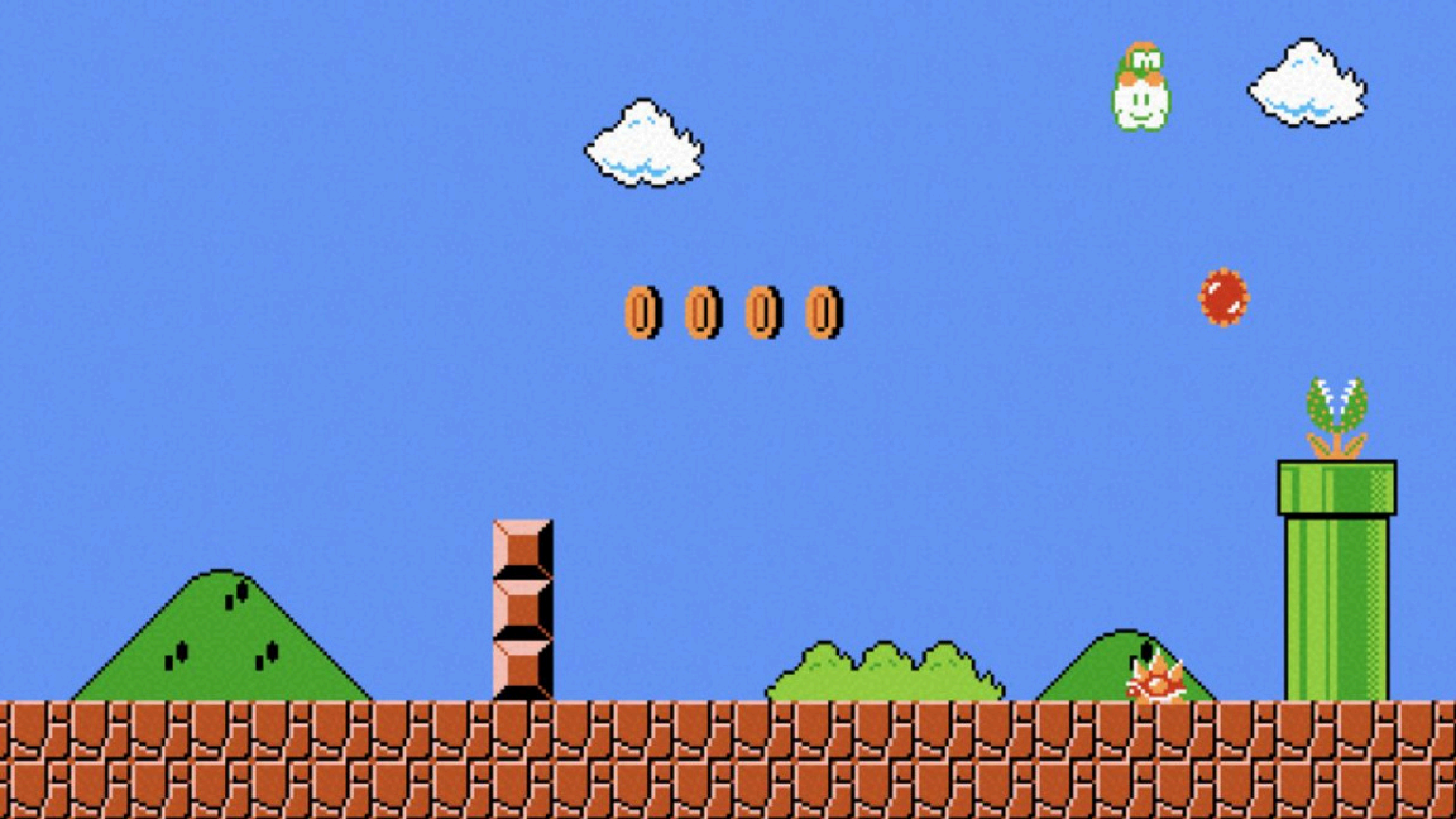

```
while (true)
{

}
```

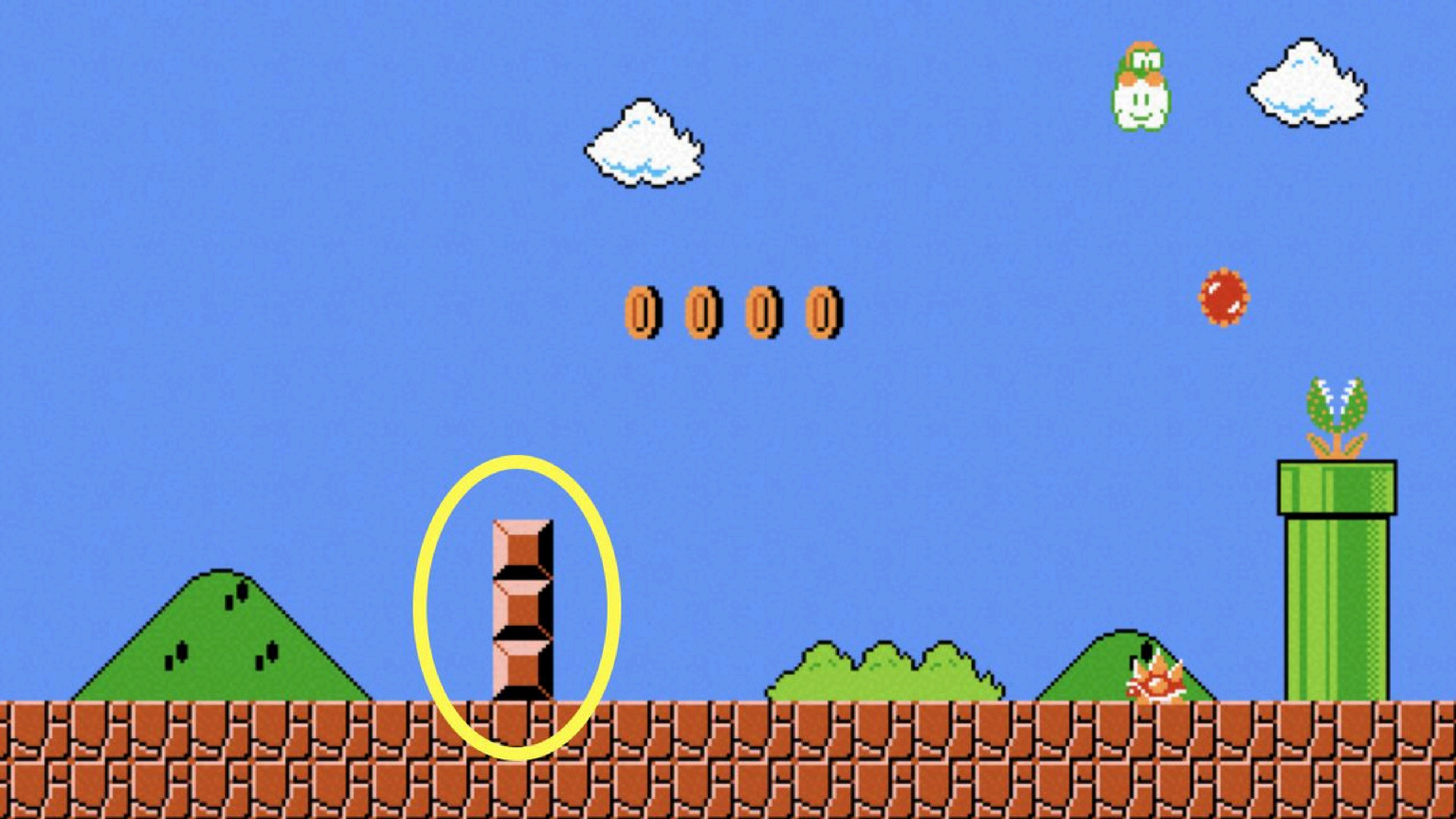


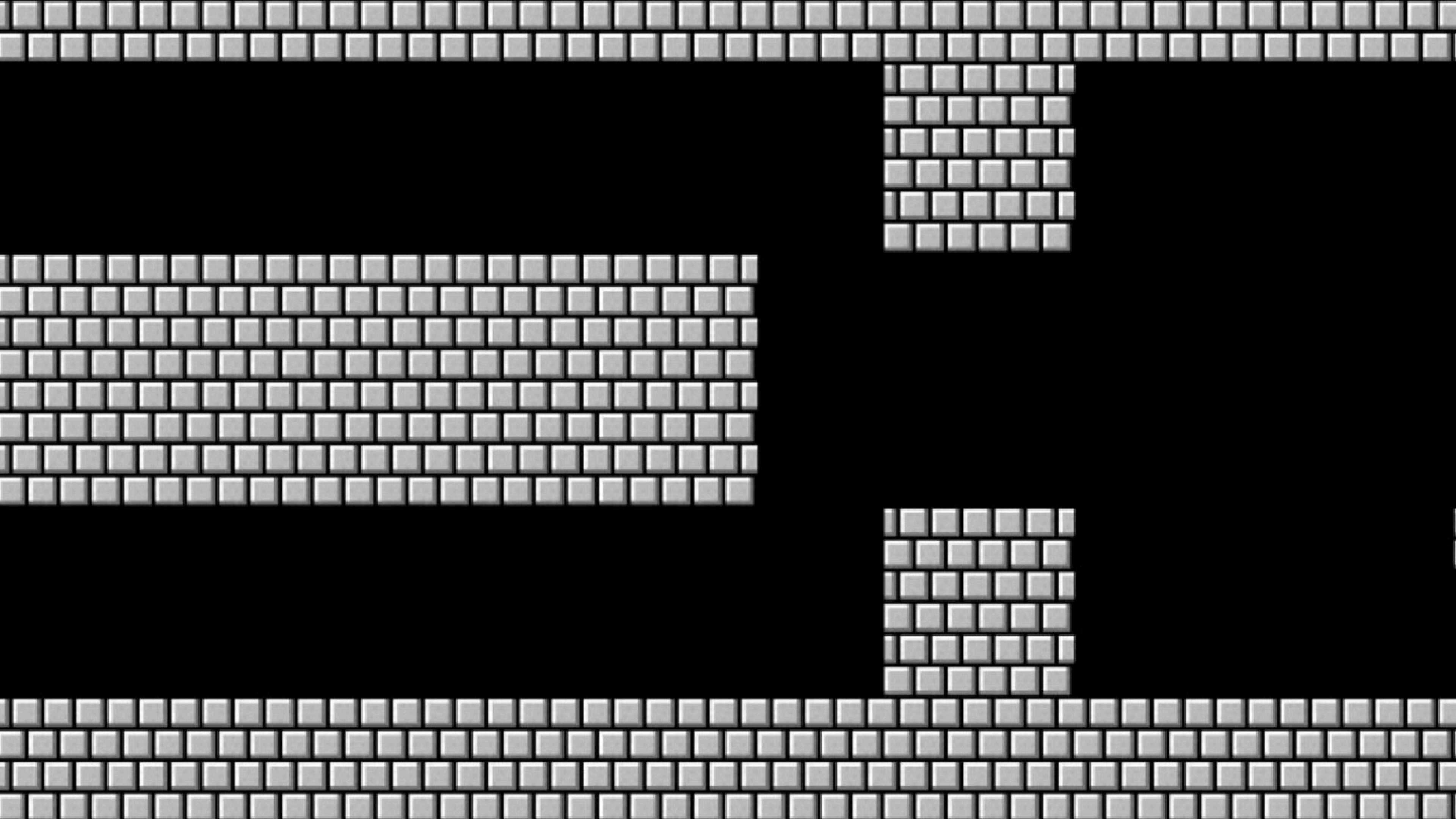

```
while (true)
{
    printf("meow\n");
}
```

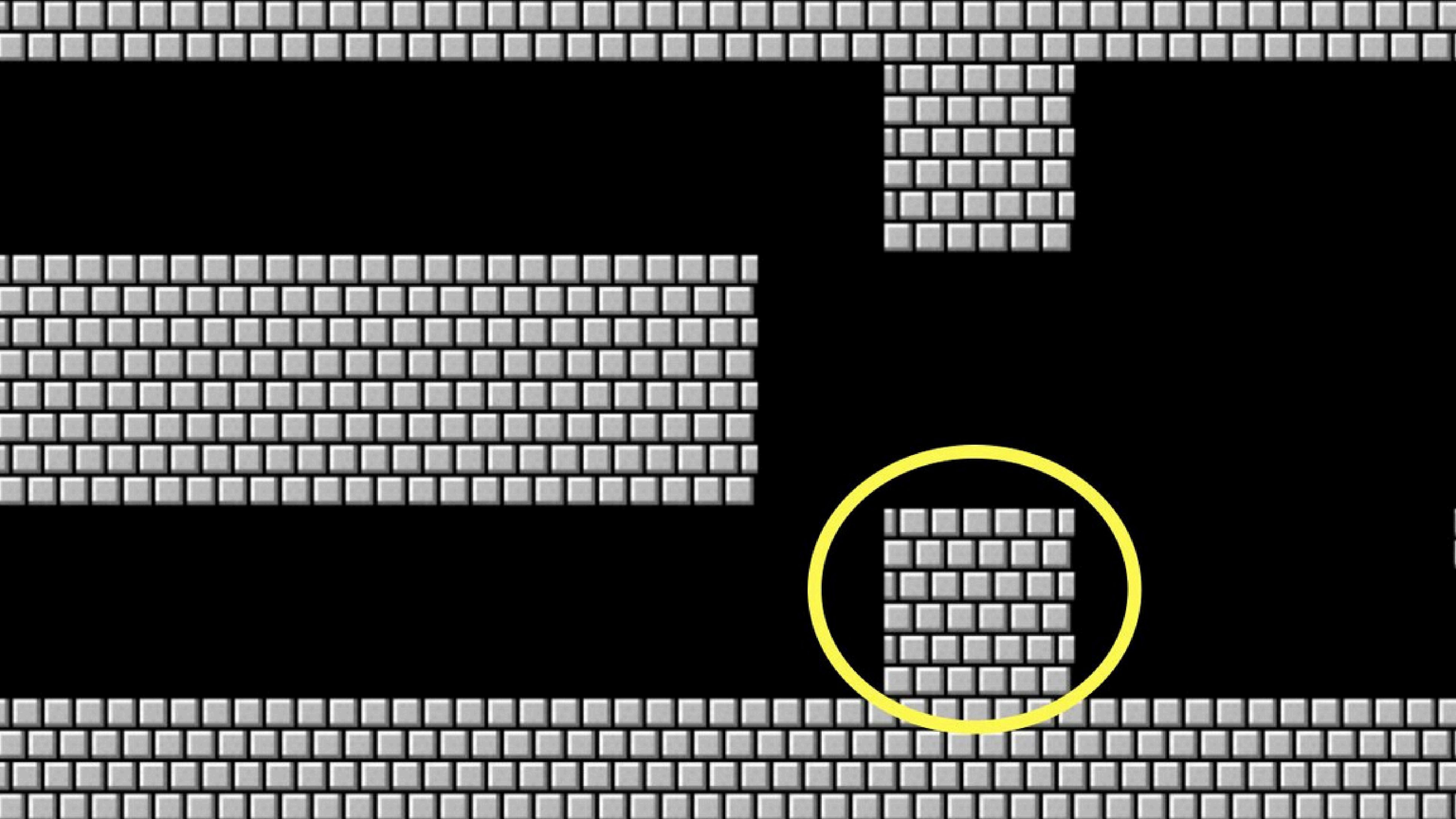

while (not edge) {
run();
}
do {
run();
} while (not edge);


FPS : 46.04 , RFPS : 46.04
MARIO
000000
×00
WORLD
1-1
TIME
SUPER
MARIO BROS.
©1985 NINTENDO
1 PLAYER GAME
2 PLAYER GAME
TOP- 000000









integer overflow


KN-13
94V-0
crucial
by Micron
47J6
4G85
8BBI2
D9HXT
4G85
8BBI2
D9HXT
4G85

000

001

010

011

100

101

110

111

1000

000

Using MAME to warp to level 256, the split screen is shown.

truncation

imprecisão de ponto-flutuante

floating-point imprecision

Year 2000 Problem

bug do milênio

1999

1999

1900

Year 2038 Problem

01 de Janeiro de 1970

19 de Janeiro de 2038

13 de Dezembro de 1901

19 de Janeiro de 2038